The Algebraic Approach to the Quantum Harmonic Oscillator as a Foundation for Quantum Field Theory

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The quantum harmonic oscillator serves as much more than a simple model for molecular vibrations; it is the fundamental building block of quantum field theory (QFT). Because any smooth potential can be approximated as a harmonic oscillator near its stable minimum, and because free quantum fields are mathematically equivalent to infinite sets of independent harmonic oscillators, mastering this system is essential for students aiming to understand one of the foundations of modern theoretical physics.

In this supplementary note, we present a rigorous, step-by-step derivation of the 1-dimensional quantum harmonic oscillator using the algebraic method. Unlike standard textbook treatments that introduce ladder operators without motivation, we physically motivate their definitions by directly factoring the quantum Hamiltonian. We then transition into Dirac notation to define the number operator, prove the discrete energy spectrum $E_n = \hbar\omega(n + 1/2)$, and normalize the state vectors. We bridge this abstract formalism with position-space wavefunctions by explicitly deriving the ground state, and we demonstrate the physical origin of the zero-point energy via the Uncertainty Principle. The reader is assumed to have a working familiarity with standard quantum mechanics, such as the time-independent Schrödinger equation, canonical commutation relations, and Dirac notation.

I. INTRODUCTION

The harmonic oscillator is one of the most important physical models in all of physics. Classically, it describes any system subject to a linear restoring force, allowing an infinite

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and continuous spectrum of allowed energies [1]. However, at the quantum level, particles confined in a potential well can only possess discrete, quantized energy states [2].

One can solve the quantum harmonic oscillator by applying the power series method to the Schrödinger equation, a process that leads to Hermite polynomials [2]. Alternatively, the algebraic approach pioneered by P.A.M. Dirac [3] and formalized in modern texts [4] provides a much simpler and more elegant method for finding the energy eigenvalues and state vectors. However, standard textbooks often introduce the algebraic "ladder" operators as a mathematical trick pulled out of thin air, leaving students wondering about their origin. In this supplementary note, we will instead build these operators naturally from the ground up by attempting to factor the Hamiltonian. This method not only simplifies calculations but also sets the stage for Quantum Field Theory (QFT), where these exact ladder operators are promoted to create and annihilate fundamental particles [5].

This document is organized as follows. In Section II, we transition from the classical to the quantum Hamiltonian. In Section III, we motivate the creation and annihilation operators by factoring the Hamiltonian. Section IV establishes the commutation relations and proves the discrete energy spectrum. Section V explicitly normalizes the matrix elements of the ladder operators using Dirac notation. Section VI derives the position-space wavefunctions. Finally, Section VII calculates expectation values to predict the zero-point energy, and Section VIII concludes by bridging this formalism into QFT.

II. FROM CLASSICAL TO QUANTUM HAMILTONIAN

For the classical harmonic oscillator, the Hamiltonian (total energy) is given by:

$$H(x, p_x) = \frac{p_x^2}{2m} + \frac{1}{2}kx^2 \tag{1}$$

where k is the force constant. For the classical harmonic oscillator, there is an infinite, continuous number of allowed energies for the system.

In quantum mechanics, momentum and position become operators via the standard quantum substitution, which in 1D is:

$$p_x \to \hat{p}_x = -i\hbar\frac{d}{dx} \quad \text{and} \quad x \to \hat{x} = x \tag{2}$$

Additionally, we usually express the force constant k in terms of the mass m and angular frequency ω because ω is a quantity that is directly observable in experiments (e.g., via spectroscopy and energy transitions $\Delta E = \hbar\omega$). Hence, the quantum Hamiltonian is:

$$\begin{aligned}\hat{H} &= \frac{\hat{p}_x^2}{2m} + \frac{1}{2}m\omega^2\hat{x}^2 \\ &= \frac{1}{2m}\left[\hat{p}_x^2 + (m\omega\hat{x})^2\right]\end{aligned} \tag{3}$$

The time-independent Schrödinger equation for the quantum harmonic oscillator is then:

$$\frac{1}{2m}\left[\hat{p}_x^2 + (m\omega\hat{x})^2\right]\psi = E\psi \tag{4}$$

III. FACTORING THE HAMILTONIAN: MOTIVATING THE LADDER OPERATORS

To solve this algebraically, we seek to "factor" the Hamiltonian. Notice that form $\hat{p}_x^2 + (m\omega\hat{x})^2$ resembles a sum of squares, $a^2 + b^2 = (a - ib)(a + ib)$. Hence, let us define two operators based on this classical factoring:

$$\hat{A}^\dagger = \frac{1}{\sqrt{2m}}(-i\hat{p}_x + m\omega\hat{x}), \quad \hat{A} = \frac{1}{\sqrt{2m}}(i\hat{p}_x + m\omega\hat{x}) \tag{5}$$

Before proceeding, let us verify the Hermitian conjugate relationship. Since position and momentum correspond to physical observables, they are Hermitian operators ($\hat{x}^\dagger = \hat{x}$ and $\hat{p}_x^\dagger = \hat{p}_x$). Taking the adjoint of $\hat{A}$ yields:

$$\hat{A}^\dagger = \frac{1}{\sqrt{2m}}(i\hat{p}_x + m\omega\hat{x})^\dagger = \frac{1}{\sqrt{2m}}(-i\hat{p}_x + m\omega\hat{x}) \tag{6}$$

which perfectly matches our definition, confirming they are adjoints.

We now calculate their product to see if we recover the Hamiltonian:

$$\begin{aligned}\hat{A}^\dagger\hat{A} &= \frac{1}{2m}(-i\hat{p}_x + m\omega\hat{x})(i\hat{p}_x + m\omega\hat{x}) \\ &= \frac{1}{2m}\left[\hat{p}_x^2 + (m\omega\hat{x})^2 + im\omega\hat{x}\hat{p}_x - im\omega\hat{p}_x\hat{x}\right] \\ &= \frac{1}{2m}\left[\hat{p}_x^2 + (m\omega\hat{x})^2 + im\omega(\hat{x}\hat{p}_x - \hat{p}_x\hat{x})\right]\end{aligned}$$

Notice exactly *why* the classical factoring fails to yield just $\hat{H}$. Because $\hat{x}$ and $\hat{p}_x$ do not commute, the $(\hat{x}\hat{p}_x - \hat{p}_x\hat{x})$ term do not cancel out to zero as they would for ordinary numbers. This non-cancellation is the mathematical origin of the zero-point energy. Using the canonical commutation relation, $[\hat{x}, \hat{p}_x] = \hat{x}\hat{p}_x - \hat{p}_x\hat{x} = i\hbar$, we get:

$$\hat{A}^\dagger\hat{A} = \frac{1}{2m}\left[\hat{p}_x^2 + (m\omega\hat{x})^2 - \hbar m\omega\right]$$

$$= \hat{H} - \frac{\hbar\omega}{2}$$

Thus, the Hamiltonian can be written as:

$$\hat{H} = \hat{A}^\dagger \hat{A} + \frac{\hbar\omega}{2} \tag{7}$$

Because $\hat{A}^\dagger \hat{A}$ has units of energy, it is highly advantageous to factor out the natural energy scale of the oscillator, $\hbar\omega$:

$$\hat{H} = \hbar\omega \left(\frac{\hat{A}^\dagger \hat{A}}{\hbar\omega} + \frac{1}{2} \right) \tag{8}$$

By absorbing the $\hbar\omega$ into the operators, we naturally arrive at the standard dimensionless creation ($\hat{a}^\dagger$) and annihilation ($\hat{a}$) operators found in textbooks, resolving the mystery of their normalization constants:

$$\boxed{\hat{H} = \hbar\omega \left(\hat{a}^\dagger \hat{a} + \frac{1}{2} \right)} \tag{9}$$

where:

$$\boxed{\hat{a}^\dagger = \frac{\hat{A}^\dagger}{\sqrt{\hbar\omega}} = \frac{1}{\sqrt{2\hbar m\omega}}(-i\hat{p}_x + m\omega\hat{x}), \quad \hat{a} = \frac{\hat{A}}{\sqrt{\hbar\omega}} = \frac{1}{\sqrt{2\hbar m\omega}}(i\hat{p}_x + m\omega\hat{x})} \tag{10}$$

We define the dimensionless **number operator** as $\hat{n} = \hat{a}^\dagger \hat{a}$, allowing us to express the Hamiltonian as

$$\hat{H} = \hbar\omega \left(\hat{n} + \frac{1}{2} \right) \tag{11}$$

As we will see, the name "number operator" comes from the fact that its eigenvalues are the non-negative integers ($n = 0, 1, 2, \dots$), which physically count the number of energy quanta present in the oscillator.

IV. THE ENERGY SPECTRUM AND COMMUTATION RELATIONS

To understand how these operators affect the quantum state, we first need their commutation relation. Expanding $\hat{a}\hat{a}^\dagger$ yields:

$$\begin{aligned} \hat{a}\hat{a}^\dagger &= \frac{1}{2\hbar m\omega}(i\hat{p}_x + m\omega\hat{x})(-i\hat{p}_x + m\omega\hat{x}) \\ &= \frac{1}{2\hbar m\omega}\left[\hat{p}_x^2 + (m\omega\hat{x})^2 + \hbar m\omega\right] \\ &= \frac{1}{\hbar\omega}\left[\frac{\hat{p}_x^2}{2m} + \frac{1}{2}m\omega\hat{x}^2\right] + \frac{1}{2} \end{aligned}$$

$$\hat{a}\hat{a}^\dagger = \frac{\hat{H}}{\hbar\omega} + \frac{1}{2}$$

From eq. (9), we know $\hat{a}^\dagger\hat{a} = \frac{\hat{H}}{\hbar\omega} - \frac{1}{2}$. Thus, the commutator is:

$$\begin{aligned}
[\hat{a}^\dagger, \hat{a}] &= \hat{a}^\dagger\hat{a} - \hat{a}\hat{a}^\dagger \\
&= \left(\frac{\hat{H}}{\hbar\omega} - \frac{1}{2}\right) - \left(\frac{\hat{H}}{\hbar\omega} + \frac{1}{2}\right) \\
[\hat{a}^\dagger, \hat{a}] &= -1
\end{aligned}$$

which, equivalently means $[\hat{a}, \hat{a}^\dagger] = 1$.

Let a quantum state with energy E be denoted by $|\psi\rangle$.

Claim 1

If $\hat{H}|\psi\rangle = E|\psi\rangle$, then $\hat{H}(\hat{a}^\dagger|\psi\rangle) = (E + \hbar\omega)(\hat{a}^\dagger|\psi\rangle)$.

Proof:

$$\begin{aligned}
\hat{H}(\hat{a}^\dagger|\psi\rangle) &= \hbar\omega\left(\hat{a}^\dagger\hat{a} + \frac{1}{2}\right)(\hat{a}^\dagger|\psi\rangle) \\
&= \hbar\omega\left(\hat{a}^\dagger\hat{a}^\dagger\hat{a}^\dagger + \frac{1}{2}\hat{a}^\dagger\right)|\psi\rangle \\
\hat{H}(\hat{a}^\dagger|\psi\rangle) &= \hat{a}^\dagger\left[\hbar\omega\left(\hat{a}\hat{a}^\dagger + \frac{1}{2}\right)\right]|\psi\rangle
\end{aligned}$$

where we factored $\hat{a}^\dagger$ to the left using the associativity of operators, $\hat{a}^\dagger\hat{a}\hat{a}^\dagger = \hat{a}^\dagger(\hat{a}\hat{a}^\dagger)$.

Using the commutator $[\hat{a}^\dagger, \hat{a}] = -1 \implies \hat{a}\hat{a}^\dagger = \hat{a}^\dagger\hat{a} + 1$, we have:

$$\begin{aligned}
\hat{H}(\hat{a}^\dagger|\psi\rangle) &= \hat{a}^\dagger\left[\hbar\omega\left(\hat{a}^\dagger\hat{a} + 1 + \frac{1}{2}\right)\right]|\psi\rangle \\
&= \hat{a}^\dagger\left[\hbar\omega\left(\hat{a}^\dagger\hat{a} + \frac{1}{2}\right) + \hbar\omega\right]|\psi\rangle \\
&= \hat{a}^\dagger\left[\hat{H} + \hbar\omega\right]|\psi\rangle \\
&= \hat{a}^\dagger(\hat{H}|\psi\rangle) + \hbar\omega(\hat{a}^\dagger|\psi\rangle) \\
&= \hat{a}^\dagger(E|\psi\rangle) + \hbar\omega(\hat{a}^\dagger|\psi\rangle) \\
\hat{H}(\hat{a}^\dagger|\psi\rangle) &= (E + \hbar\omega)(\hat{a}^\dagger|\psi\rangle)
\end{aligned}$$

Thus, $\hat{a}^\dagger$ acts as a “raising” operator, raising the energy up by $\hbar\omega$.

Claim 2

If $\hat{H}\,|\psi\rangle = E\,|\psi\rangle$, then $\hat{H}(\hat{a}\,|\psi\rangle) = (E - \hbar\omega)(\hat{a}\,|\psi\rangle)$.

Proof:

$$\begin{aligned} \hat{H}(\hat{a}\,|\psi\rangle) &= \hbar\omega\left(\hat{a}^\dagger\hat{a} + \frac{1}{2}\right)(\hat{a}\,|\psi\rangle) \\ &= \left[\hbar\omega\left(\hat{a}^\dagger\hat{a}\hat{a} + \frac{1}{2}\hat{a}\right)\right]|\psi\rangle \end{aligned}$$

Using $[\hat{a}^\dagger, \hat{a}] = -1 \implies \hat{a}^\dagger\hat{a} = \hat{a}\hat{a}^\dagger - 1$, we substitute into the first term:

$$\begin{aligned} \hat{H}(\hat{a}\,|\psi\rangle) &= \left[\hbar\omega\left((\hat{a}\hat{a}^\dagger - 1)\hat{a} + \frac{1}{2}\hat{a}\right)\right]|\psi\rangle \\ &= \hat{a}\left[\hbar\omega\left(\hat{a}^\dagger\hat{a} - 1 + \frac{1}{2}\right)\right]|\psi\rangle \\ &= \hat{a}\left[\hbar\omega\left(\hat{a}^\dagger\hat{a} + \frac{1}{2}\right) - \hbar\omega\right]|\psi\rangle \\ &= \hat{a}(\hat{H} - \hbar\omega)\,|\psi\rangle \\ &= \hat{a}(E\,|\psi\rangle) - \hbar\omega(\hat{a}\,|\psi\rangle) \\ \hat{H}(\hat{a}\,|\psi\rangle) &= (E - \hbar\omega)(\hat{a}\,|\psi\rangle) \end{aligned}$$

Thus, $\hat{a}$ acts as a “lowering” operator.

Claim 3

The energy spectrum of the quantum harmonic oscillator is bounded from below, with energy levels given by $E_n = \hbar\omega\left(n + \frac{1}{2}\right)$, where $n \in \mathbb{Z}_{\geq 0}$.

Proof: Energy in a stable potential well cannot become infinitely negative. We prove this rigorously by taking the expectation value of $\hat{H}$ for any state $|\psi\rangle$:

$$\begin{aligned} \langle\psi|\hat{H}|\psi\rangle &= \langle\psi|\hbar\omega\left(\hat{a}^\dagger\hat{a} + \tfrac{1}{2}\right)|\psi\rangle \\ &= \hbar\omega\,\langle\psi|\hat{a}^\dagger\hat{a}|\psi\rangle + \frac{\hbar\omega}{2}\,\langle\psi|\psi\rangle \\ \langle\psi|\hat{H}|\psi\rangle &= \hbar\omega\,\langle\psi|\hat{a}^\dagger\hat{a}|\psi\rangle + \frac{\hbar\omega}{2} \end{aligned}$$

Because $\langle\psi|\hat{a}^\dagger\hat{a}|\psi\rangle$ is the squared norm of $\hat{a}\,|\psi\rangle$, it must be ≥ 0, so $\langle\psi|\hat{H}|\psi\rangle \geq \hbar\omega/2$.

To prevent the lowering operator from producing a state with negative energy, there must exist a ground state $|0\rangle$ satisfying:

$$\hat{a}\,|0\rangle = 0 \tag{12}$$

The ground state energy follows directly:

$$\hat{H}|0\rangle = \hbar\omega\big(\hat{a}^\dagger\hat{a} + \tfrac{1}{2}\big)|0\rangle$$

$$E_0|0\rangle = \frac{\hbar\omega}{2}|0\rangle \quad \Rightarrow \quad \boxed{E_0 = \frac{\hbar\omega}{2}}$$

Since each application of $\hat{a}^\dagger$ raises the energy by $\hbar\omega$, the n-th energy eigenvalue is $E_n = E_0 + n\hbar\omega$, giving the quantized spectrum:

$$\boxed{E_n = \hbar\omega\big(n + \tfrac{1}{2}\big)} \tag{13}$$

Claim 4

The states $|n\rangle$ are eigenstates of the number operator such that $\hat{N}|n\rangle = n|n\rangle$.

Proof: We proceed by induction. For the ground state, $\hat{a}|0\rangle = 0$, so $\hat{n}|0\rangle = \hat{a}^\dagger\hat{a}|0\rangle = 0 = 0|0\rangle$; the base case holds. Assume the claim holds for $|n\rangle$. For $\hat{a}^\dagger|n\rangle$, we apply the commutator $[\hat{n}, \hat{a}^\dagger] = \hat{a}^\dagger$:

$$\begin{aligned}
\hat{n}(\hat{a}^\dagger|n\rangle) &= (\hat{a}^\dagger\hat{n} + \hat{a}^\dagger)|n\rangle \\
&= \hat{a}^\dagger(n|n\rangle) + \hat{a}^\dagger|n\rangle \\
\hat{n}(\hat{a}^\dagger|n\rangle) &= (n+1)(\hat{a}^\dagger|n\rangle)
\end{aligned}$$

Thus each application of the raising operator increases the eigenvalue of $\hat{n}$ by exactly 1, proving $\hat{n}|n\rangle = n|n\rangle$ for all $n \in \mathbb{Z}_{\geq 0}$.

V. NORMALIZATION AND THE GENERAL STATE FORMULA

While we proved that applying $\hat{a}^\dagger$ to state $|n\rangle$ yields a state proportional to $|n+1\rangle$, we must find the exact normalization constants. Let $\hat{a}^\dagger|n\rangle = c_n|n+1\rangle$. Demanding that the new state is normalized, i.e., $\langle n+1|n+1\rangle = 1$, we take the inner product:

$$\begin{aligned}
|c_n|^2 &= (\langle n|\hat{a})(\hat{a}^\dagger|n\rangle) \\
&= \langle n|\hat{a}\hat{a}^\dagger|n\rangle \\
&= \langle n|(\hat{n}+1)|n\rangle \quad \text{(using } [\hat{a}, \hat{a}^\dagger] = 1\text{)} \\
&= n + 1
\end{aligned}$$

Choosing the real positive root, $c_n = \sqrt{n+1}$.

Similarly, for the lowering operator, let $\hat{a}\left|n\right\rangle = d_n \left|n-1\right\rangle$. Taking the inner product:

$$\begin{aligned} |d_n|^2 &= (\left\langle n\right| \hat{a}^\dagger)(\hat{a}\left|n\right\rangle) \\ &= \left\langle n\right| \hat{a}^\dagger \hat{a} \left|n\right\rangle \\ &= \left\langle n\right| \hat{n} \left|n\right\rangle \\ &= n \end{aligned}$$

Choosing the real positive root yields $d_n = \sqrt{n}$. This gives the vital matrix element formulas:

$$\boxed{\hat{a}^\dagger \left|n\right\rangle = \sqrt{n+1}\left|n+1\right\rangle, \quad \hat{a}\left|n\right\rangle = \sqrt{n}\left|n-1\right\rangle} \tag{14}$$

With the raising operator matrix element established, we can construct any arbitrary excited state $\left|n\right\rangle$ directly from the ground state $\left|0\right\rangle$. Applying $\hat{a}^\dagger$ repeatedly:

$$\begin{aligned} \left|1\right\rangle &= \hat{a}^\dagger \left|0\right\rangle \\ \left|2\right\rangle &= \frac{1}{\sqrt{2}}\hat{a}^\dagger \left|1\right\rangle = \frac{1}{\sqrt{2\cdot 1}}(\hat{a}^\dagger)^2 \left|0\right\rangle \\ \left|3\right\rangle &= \frac{1}{\sqrt{3}}\hat{a}^\dagger \left|2\right\rangle = \frac{1}{\sqrt{3\cdot 2\cdot 1}}(\hat{a}^\dagger)^3 \left|0\right\rangle \end{aligned}$$

From the pattern, we can see that this naturally yields the general state construction formula:

$$\boxed{\left|n\right\rangle = \frac{(\hat{a}^\dagger)^n}{\sqrt{n!}}\left|0\right\rangle} \tag{15}$$

This purely algebraic construction of the state space from the vacuum is a hallmark of modern quantum mechanics [4], and foreshadows the space structure utilized in QFT [5].

VI. POSITION-SPACE WAVEFUNCTIONS

To map the abstract state ground $\left|0\right\rangle$ to a position-space wavefunction $\psi_0(x) = \langle x|0\rangle$, we project the annihilation condition $\hat{a}\left|0\right\rangle = 0$ into position space:

$$\begin{aligned} \frac{1}{\sqrt{2\hbar m\omega}}(i\hat{p}_x + m\omega\hat{x})\psi_0 = 0 \quad &\Longrightarrow \quad \left[i\left(-i\hbar\frac{d}{dx}\right) + m\omega x\right]\psi_0 = 0 \\ &\Longrightarrow \quad \hbar\frac{d\psi_0}{dx} = -m\omega x\,\psi_0 \\ &\Longrightarrow \quad \frac{d\psi_0}{\psi_0} = -\frac{m\omega}{\hbar}x\,dx \end{aligned}$$

Integrating both sides, we get:

$$\ln|\psi_0| = -\frac{m\omega}{2\hbar}x^2 + A' \quad \Longrightarrow \quad \psi_0 = Ae^{-\frac{m\omega}{2\hbar}x^2}$$

We normalize the wavefunction over all space:

$$\int_{-\infty}^{\infty} |\psi_0|^2 dx = |A|^2 \int_{-\infty}^{\infty} e^{-\frac{m\omega}{\hbar}x^2} dx$$
$$1 = |A|^2 \sqrt{\frac{\pi\hbar}{m\omega}} \quad \Longrightarrow \quad A = \left(\frac{m\omega}{\pi\hbar}\right)^{1/4}$$

Thus, the normalized ground state wavefunction is:

$$\boxed{\psi_0(x) = \left(\frac{m\omega}{\pi\hbar}\right)^{1/4} e^{-\frac{m\omega}{2\hbar}x^2}} \tag{16}$$

To find the first excited state, we simply apply the position-space representation of the creation operator to $\psi_0(x)$, utilizing $|1\rangle = \hat{a}^\dagger |0\rangle$:

$$\begin{aligned}\psi_1(x) &= \frac{1}{\sqrt{2\hbar m\omega}} \left(m\omega x - \hbar\frac{d}{dx}\right) \left(\frac{m\omega}{\pi\hbar}\right)^{1/4} e^{-\frac{m\omega}{2\hbar}x^2} \\ &= \frac{1}{\sqrt{2\hbar m\omega}} (m\omega x + m\omega x)\, \psi_0(x) \\ &= \sqrt{\frac{2m\omega}{\hbar}} x\, \psi_0(x)\end{aligned}$$

This elegantly demonstrates how the algebraic method automatically generates the Hermite polynomials.

VII. THE ZERO-POINT ENERGY AND THE UNCERTAINTY PRINCIPLE

The true power of the algebraic method is that it eliminates the need to evaluate integrals of Hermite polynomials to find expectation values. By adding and subtracting the two equations in eq. (10), we can express position and momentum strictly in terms of the ladder operators:

$$\hat{x} = \sqrt{\frac{\hbar}{2m\omega}}(\hat{a} + \hat{a}^\dagger), \quad \hat{p}_x = -i\sqrt{\frac{\hbar m\omega}{2}}(\hat{a} - \hat{a}^\dagger) \tag{17}$$

Consider the ground state $|0\rangle$. The expectation value of position is trivial because ladder operators map $|0\rangle$ to orthogonal states:

$$\langle 0|\hat{x}|0\rangle \propto \langle 0|\,(\hat{a} + \hat{a}^\dagger)\,|0\rangle = 0 + 0 = 0 \tag{18}$$

By the same logic, $\langle 0|\hat{p}_x|0\rangle = 0$. However, the expectation value of $\hat{x}^2$ is non-zero. Expanding $(\hat{a}+\hat{a}^\dagger)^2$:

$$\begin{aligned}
\langle 0|\hat{x}^2|0\rangle &= \frac{\hbar}{2m\omega}\langle 0|\left(\hat{a}\hat{a}+\hat{a}^\dagger\hat{a}^\dagger+\hat{a}^\dagger\hat{a}+\hat{a}\hat{a}^\dagger\right)|0\rangle \\
&= \frac{\hbar}{2m\omega}\langle 0|\left(\hat{a}\hat{a}^\dagger\right)|0\rangle \quad \text{(since all others annihilate against } |0\rangle \text{ or } \langle 0|\text{)} \\
&= \frac{\hbar}{2m\omega}\langle 0|\left(\hat{N}+1\right)|0\rangle = \frac{\hbar}{2m\omega}
\end{aligned}$$

Similarly, evaluating $\langle 0|\hat{p}_x^2|0\rangle$ yields $\frac{\hbar m\omega}{2}$. We can now find the exact uncertainties in the ground state:

$$\Delta x = \sqrt{\langle \hat{x}^2\rangle - \langle \hat{x}\rangle^2} = \sqrt{\frac{\hbar}{2m\omega}}, \quad \Delta p_x = \sqrt{\langle \hat{p}_x^2\rangle - \langle \hat{p}_x\rangle^2} = \sqrt{\frac{\hbar m\omega}{2}} \tag{19}$$

Multiplying them together gives $\Delta x \Delta p_x = \hbar/2$.

We can push this realization even further. Instead of just verifying the bound, we can *predict* the zero-point energy strictly from first principles. By symmetry, $\langle \hat{x}\rangle = \langle \hat{p}_x\rangle = 0$, so $\langle \hat{x}^2\rangle = (\Delta x)^2$ and $\langle \hat{p}_x^2\rangle = (\Delta p_x)^2$. The total expected energy is:

$$E = \left\langle \hat{H} \right\rangle = \frac{(\Delta p_x)^2}{2m} + \frac{1}{2}m\omega^2(\Delta x)^2 \tag{20}$$

From the uncertainty principle, we know $(\Delta p_x)^2 \geq \frac{\hbar^2}{4(\Delta x)^2}$. Substituting this constraint to minimize E:

$$E \geq \frac{\hbar^2}{8m(\Delta x)^2} + \frac{1}{2}m\omega^2(\Delta x)^2 \tag{21}$$

To find the minimum possible energy allowed by nature, we take the derivative with respect to $(\Delta x)^2$ and set it to zero:

$$\frac{dE}{d(\Delta x)^2} = -\frac{\hbar^2}{8m(\Delta x)^4} + \frac{1}{2}m\omega^2 = 0 \quad \Longrightarrow \quad (\Delta x)^2 = \frac{\hbar}{2m\omega} \tag{22}$$

Plugging this minimizing variance back into our energy inequality yields:

$$E_{min} = \frac{\hbar^2}{8m}\left(\frac{2m\omega}{\hbar}\right) + \frac{1}{2}m\omega^2\left(\frac{\hbar}{2m\omega}\right) = \frac{\hbar\omega}{4} + \frac{\hbar\omega}{4} = \frac{\hbar\omega}{2} \tag{23}$$

The ground state energy $E_0 = \hbar\omega/2$ is identically the absolute minimum energy required to satisfy the Heisenberg Uncertainty Principle.

Classically, the lowest possible energy occurs when the particle is stationary and at the origin ($x = 0$, $p = 0$). But quantum mechanically, the uncertainty principle limits our precision of x and p. That is, you cannot have $x = 0$ and $p = 0$ at the same time. Even in the lowest energy state, motion will still happen such that $p \neq 0$, which means the lowest energy is not zero, in contrast to classical mechanics.

VIII. CONCLUSION AND CONNECTION TO QFT

In this supplementary note, we derived the 1-dimensional quantum harmonic oscillator from first principles. By physically factoring the Hamiltonian, we naturally motivated the definitions of the creation ($\hat{a}^\dagger$) and annihilation ($\hat{a}$) operators. Transitioning to Dirac notation allowed us to bypass the heavy machinery of solving second-order differential equations, relying instead on canonical commutation relations to map the discrete energy spectrum and construct expectation values algebraically.

The true culmination of this framework lies beyond single-particle mechanics. In Quantum Field Theory (QFT), continuous fields (like the electromagnetic field) are mathematically Fourier-decomposed into infinite sets of independent momentum modes [5]. Remarkably, the Hamiltonian for each individual mode takes the exact algebraic form of the harmonic oscillator. Consequently, the operators $\hat{a}^\dagger$ and $\hat{a}$ take on a profound physical meaning: they no longer shift the energy state of a single confined particle, but rather they *create* and *annihilate* distinct particles (quanta) with specific momenta from the vacuum state.

This conceptual leap also introduces students to their first encounter with regularization. Because a field contains an infinite number of independent momentum modes (k), the total vacuum energy is the sum of all their zero-point energies: $E_{vac} = \sum_k \frac{1}{2}\hbar\omega_k$. This sum is formally divergent. To render the theory mathematically finite and physically meaningful, physicists employ a procedure called *normal ordering*, which redefines the Hamiltonian to subtract this infinite vacuum energy. This conceptually jarring but necessary technique is a direct consequence of the zero-point energy inherent to the algebraic oscillator.

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